

Deep Learning for teens

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Transfer learning vs Fine-tuning

▶ **Transfer Learning**

- ▶ Base model is frozen – none of its pretrained layers update
- ▶ Only the head trains – the layers you add on top
- ▶ Fast training – fewer parameters to update
- ▶ Lower risk of overfitting – great for small datasets

▶ **Fine-Tuning**

- ▶ Unfreeze some top layers of the base model
- ▶ Train head + part of the base model
- ▶ Very small learning rate needed
- ▶ More accuracy potential – model adapts to your dataset
- ▶ Higher risk of overfitting – especially with small datasets

Data Augmentation

- ▶ Reasons
 - ▶ Increase the amount of training data
 - ▶ Improve performance on unseen images
 - ▶ Reduce overfitting on small datasets

Types of data augmentation

Category	Augmentations
Geometric	Random flip, Random rotation, Random translation, Random zoom, Random crop, Random shear, Affine transform, Perspective transform
Color / Pixel	Random brightness, Random contrast, Random saturation, Random hue shift, Color jitter, Gaussian noise, Salt-and-pepper noise, Blur / sharpen
Occlusion	Cutout, Random erasing, GridMask
Mixing-Based	Mixup, CutMix, FMix, RICAP
Synthetic / Advanced	GAN-based augmentation, Diffusion augmentation, Style-transfer augmentation

Homework

- ▶ 1. Load the fruits-360_100x100 dataset. (/kaggle/input/fruits/fruits-360_100x100)
- ▶ 2. Check the Training and Test folders.
- ▶ 3. Resize, batch, and augment the images.
- ▶ 4. Train VGG16 with frozen layers.
- ▶ 5. Unfreeze a few layers and fine-tune.
- ▶ 6. Compare both phases and plot the results.